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Simulation of Species Interacting in a Complex Ecosystem

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Abstract

Understanding the dynamic of the food web is critical as it consists of different species in the environment that depend on each other, either directly or indirectly. This mathematical model was developed to understand better the dynamics of the ecosystem, which consists of plants, herbivores, carnivores, omnivores, scavengers, saprophages and decomposers. To understand this ecosystem, mathematical differential equations are formed with the help of various factors such as growth rates, predation rates, and intraspecific competition terms that can simulate an ecosystem with the help of graphs. Graphs such as population dynamics with time and phase diagrams between species provide information on how ecological systems work under the assumed parameters, resulting in the oscillated graph. In addition, we investigate the critical roles of scavengers, saprophages, and decomposers in nutrient recycling and their indirect effects on trophic dynamics. The model shows the mathematical application to ecological problems and its contribution to understanding population dynamics.

Keywords : Plants, Herbivores, Carnivores, Omnivores, Scavengers, Saprophages, Decomposers, Differential Equations, Trophic Dynamics.

1 Introduction

The ecosystem consists of the interactions between several species that are dependent on one another, directly or indirectly, which gives us insights into their population dynamics. A classical model called the "**Lotka–Volterra predator-prey model**" [1,2] has been used to describe this ecosystem, but using the complex interaction between species. As a result, complex food webs are produced with the interaction of species, which enables the strengthening of the energy flow and nutrient cycling. As this model is inspired by the **Lotka–Volterra predator-prey model**, it is best to know the ecological role of each species that makes up this complicated model.

Plants : They are the foundation of the ecosystem, as they are the primary producers. By converting solar energy into chemical energy through photosynthesis, they produce natural compounds that fuel higher trophic levels, such as herbivores, omnivores, saprophages, and decomposers.

Herbivores : They depend mainly on the population of plants, so the growth rate of herbivores depends on their abundance and serves as a food source for omnivores, carnivores, scavengers, and decomposers. Some examples of herbivores are cows, goats, horses, sheep and many more.

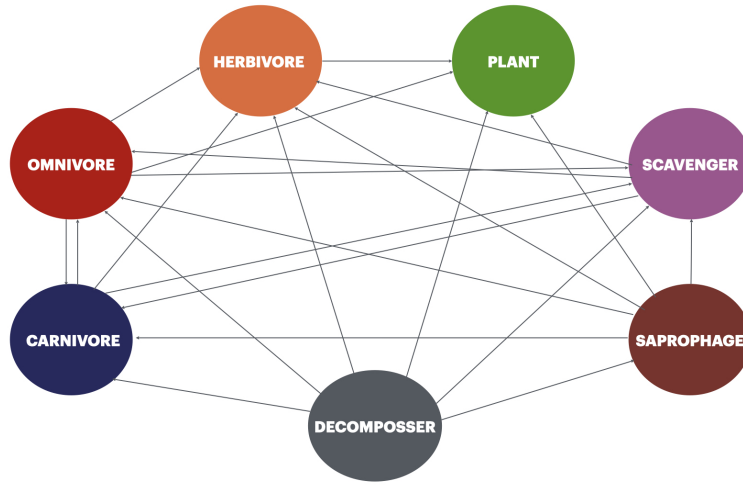


Figure 1: Food Web Visualization on Python

Omnivores : They are the type of consumer who can eat plants and animals, making them flexible species for survival. Some examples are bears, humans, birds, dogs, and many more.

Carnivores : They are the apex of the food web system as they depend on living animals. Examples are lion, tiger, leopard and many more.[7]

Scavengers : They depend mainly on dead animals, to be exact. Examples are vultures, hyenas, crows, and many more.[9]

Saprophages : They consume leftover dead animals and plants which they break down into organic matter. Examples are beetles, butterflies, flies, earthworms and many more.

Decomposers : They are the main recyclers of the environment because when species like plants and animals are decayed into organic matter, they come and break those organic matters into simpler inorganic compounds, which makes the soils enrich in minerals for plants. Growth rate.[10]

This model is designed to set a goal for a detailed environmental investigation of Web Methodology. It enables a detailed analysis of energy transfers and population dynamics.[6]

2 Method

The Lotka–Volterra model is frequently used to describe the dynamics of ecological systems where species interact, one a predator and one its prey. The model is simplified with the following assumptions: (1) only two species exist: fox and rabbit; (2) rabbits are born and then die through predation or inherent death; (3) foxes are born and their birth rate is positively affected by the rate of predation, and they die naturally. [3]

We were inspired by the Lotka–Volterra predator-prey model and created a model in which seven species interact. To implement the relationships between them, we developed formulas that describe their population rates [11,12].

2.1 Equation formation

Plants

$$\frac{dP}{dt} = pP - h_pHP - o_pOP - s_pSP - d_pDP - p_pP^2$$

Where:

- P : Plant population.
- pP : Growth rate of plants.
- h_pHP : Loss of plants due to consumption by herbivores.
- o_pOP : Loss of plants due to consumption by omnivores.
- s_pSP : Loss of plants due to consumption by saprophages.
- d_pDP : Loss of plants due to decomposition by decomposers.
- p_pP^2 : Reduction of plant growth due to intraspecific competition.

Herbivores

$$\frac{dH}{dt} = p_hPH - c_hCH - o_hOH - h_{sc}SH - h_{sa}S_aH - h_dDH - h_hH^2$$

Where:

- H : Herbivore population.
- p_hPH : Growth of herbivores due to plant consumption.
- c_hCH : Loss of herbivores due to carnivore predation.
- o_hOH : Loss of herbivores due to predation by omnivores.
- $h_{sc}SH$: Loss of herbivores due to scavengers.

- $h_{sa}S_aH$: Loss of herbivores due to saprophages.
- h_dDH : Loss of herbivores due to decomposers.
- h_hH^2 : Loss of herbivores due to intraspecific competition.

Omnivores

$$\frac{dO}{dt} = p_oPO + h_oHO + o_eCO - c_oCO - s_oSO - s_{sa}S_aO - d_oDO - o_oO^2$$

Where:

- O : Omnivore population.
- p_oPO : Growth of omnivores due to plant consumption.
- h_oHO : Growth of omnivores due to herbivore consumption.
- o_eCO : Growth of omnivores due to carnivore consumption.
- c_oCO : Loss of omnivores due to predation by carnivores.
- s_oSO : Loss of omnivores due to scavengers.
- $s_{sa}S_aO$: Loss of omnivores due to saprophages.
- d_oDO : Loss of omnivores due to decomposers.
- o_oO^2 : Loss of omnivores due to intraspecific competition.

Carnivores

$$\frac{dC}{dt} = h_cHC + o_cOC + c_sCS - o_eCO - s_cSC - s_{sa}S_aC - d_cDC - c_cC^2$$

Where:

- C : Carnivore population.
- h_cHC : Growth of carnivores due to herbivore consumption.
- o_cOC : Growth of carnivores due to omnivore consumption.
- c_sCS : Growth of carnivores due to scavenger consumption.
- o_eCO : Loss of carnivores due to predation by omnivores.
- s_cSC : Loss of carnivores due to scavengers.
- $s_{sa}S_aC$: Loss of carnivores due to saprophages.
- d_cDC : Loss of carnivores due to decomposers.
- c_cC^2 : Loss of carnivores due to intraspecific competition.

Scavengers

$$\frac{dS}{dt} = sca(H + O + C)S - o_s OS - c_s CS - s_{sa} S_a S - d_s DS - s_s S^2$$

Where:

- S : Scavenger population.
- $sca(H + O + C)S$: Growth of scavengers due to consumption of herbivores, omnivores, and carnivores.
- $o_s OS$: Loss of scavengers due to omnivores.
- $c_s CS$: Loss of scavengers due to carnivores.
- $s_{sa} S_a S$: Loss of scavengers due to saprophages.
- $d_s DS$: Loss of scavengers due to decomposers.
- $s_s S^2$: Loss of scavengers due to intraspecific competition.

Saprophages

$$\frac{dS_a}{dt} = sap(P + H + O + C + S)S_a - d_{sa} DS_a - s_{sa} S_a^2$$

Where:

- S_a : Saprophage population.
- $sap(P + H + O + C + S)S_a$: Saprophage growth due to consumption of plants, herbivores, omnivores, carnivores, and scavengers.
- $d_{sa} DS_a$: Loss of saprophages due to decomposers.
- $s_{sa} S_a^2$: Loss of saprophages due to intraspecific competition.

Decomposers

$$\frac{dD}{dt} = d(H + O + C + S + S_a)D - d_d D^2$$

Where:

- D : Decomposer population.
- $d(H + O + C + S + S_a)D$: Growth of decomposers due to the breakdown of herbivores, omnivores, carnivores, scavengers and saprophages.
- $d_d D^2$: Loss of decomposers due to intraspecific competition. [4,5]

2.2 Table of Parameters

Parameter	Description
p	Growth rate of plants
h_p	Rate at which herbivores consume plants
o_p	Rate at which omnivores consume plants
s_p	Rate at which saprophages consume plants
d_p	Rate at which decomposers consume plants
p_p	Rate of intraspecific competition among plants
p_h	Growth rate of herbivores due to plant consumption
c_h	Rate at which carnivores consume herbivores
o_h	Rate at which omnivores consume herbivores
h_{sc}	Rate at which scavengers consume herbivores
h_{sa}	Rate at which saprophages consume herbivores
h_d	Rate at which decomposers consume herbivores
h_h	Rate of intraspecific competition among herbivores
p_o	Growth rate of omnivores due to plant consumption
h_o	Growth rate of omnivores due to herbivore consumption
o_e	Growth rate of omnivores due to carnivore consumption
c_o	Rate at which carnivores consume omnivores
s_o	Rate at which scavengers consume omnivores
s_{sa}	Rate at which saprophages consume omnivores
d_o	Rate at which decomposers consume omnivores
o_o	Rate of intraspecific competition among omnivores
h_c	Growth rate of carnivores due to herbivore consumption
o_c	Growth rate of carnivores due to omnivore consumption
c_s	Growth rate of carnivores due to scavenger consumption
s_c	Rate at which scavengers consume carnivores
s_{sa}	Rate at which saprophages consume carnivores
d_c	Rate at which decomposers consume carnivores
c_c	Rate of intraspecific competition among carnivores
sca	Growth rate of scavengers
s_s	Rate of intraspecific competition among scavengers
sap	Growth rate of saprophages
d_{sa}	Rate at which decomposers consume saprophages
s_{sa}	Rate of intraspecific competition among saprophages
d	Growth rate of decomposers
d_d	Rate of intraspecific competition among decomposers

Table 1: Table of parameters and their descriptions.

2.3 Visualization method

We want to show how different species/groups in an ecosystem affect each other over time. So, we had to set the population size for every species and set the

```

#Plotting
plt.figure(figsize=(15, 60))

#Factors vs time
plt.subplot(9, 1,1)
plt.plot(t, Plants, 'green', label='Plants')
plt.plot(t, Herbivores, 'blue', label='Herbivores')
plt.plot(t, Omnivores, 'purple', label='Omnivores')
plt.plot(t, Carnivores, 'red', label='Carnivores')
plt.plot(t, Scavengers, 'brown', label='Scavengers')
plt.plot(t, Saprophages, 'cyan', label='Saprophages')
plt.plot(t, Decomposer, 'orange', label='Decomposer')
plt.xlabel('Time')
plt.ylabel('Population')
plt.title('Population Dynamics Over Time')
plt.legend()
plt.grid()

```

Figure 2: Population over Time diagram plotting code on Python

time frame. Then we were able to represent the Population dynamics over time graphs and phase diagrams by plotting them with the help of Python.

3 Results/Simulation

We successfully plotted 8 graphs in total, one of which plots the population growth rate of 7 species over time, and the other 7 are phase diagrams of the species. Now, we want to represent through a graph how various groups of species interact and influence each other's survival.[13]

3.1 Population dynamic over time

This graph represents the summation of all the elements increasing and decreasing factors, such as their growth rate, consumption rate, and intraspecific competition.

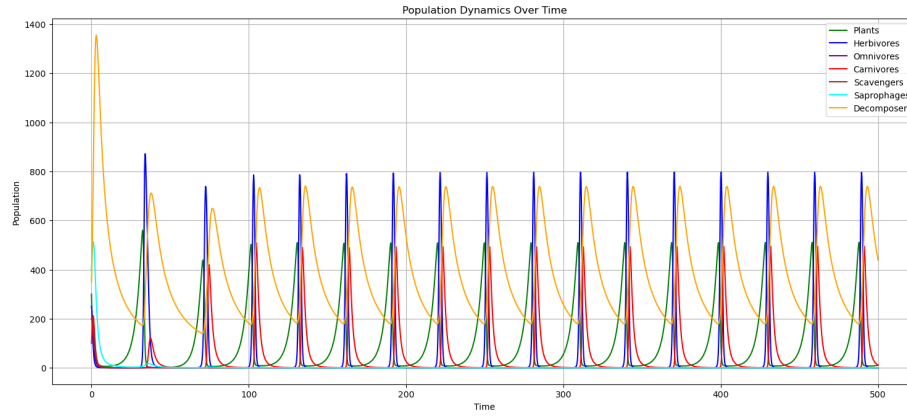


Figure 3: Population Dynamics Over Time

Here, the plants, herbivores and carnivores show a similar pattern which indicates their dependency on themselves, whereas the omnivores, scavengers and saprophages don't show similar patterns. But the patterns of decomposers are highlighted as they peaked in the population chart in the beginning and then matched the flow with Plants, herbivores and carnivores.

3.1.1 Plants vs Herbivores — Herbivores vs Carnivores

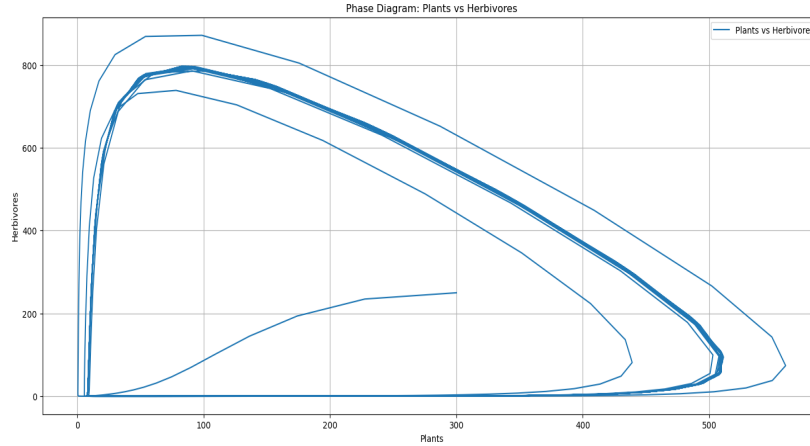


Figure 4: Phase Diagram: Plants vs Herbivores

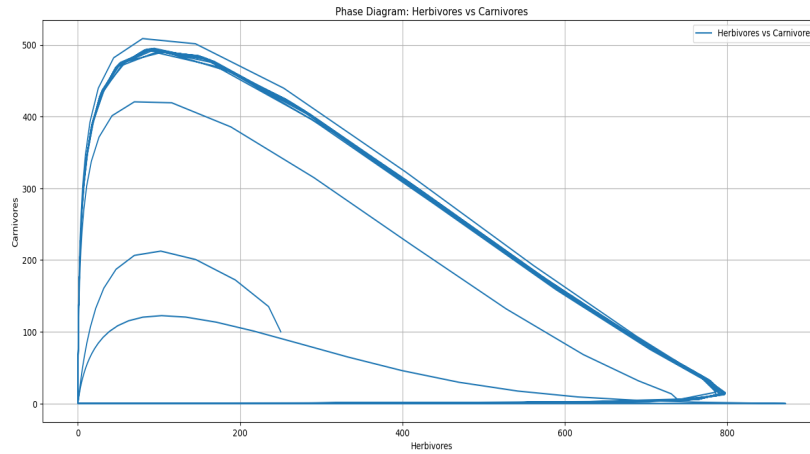
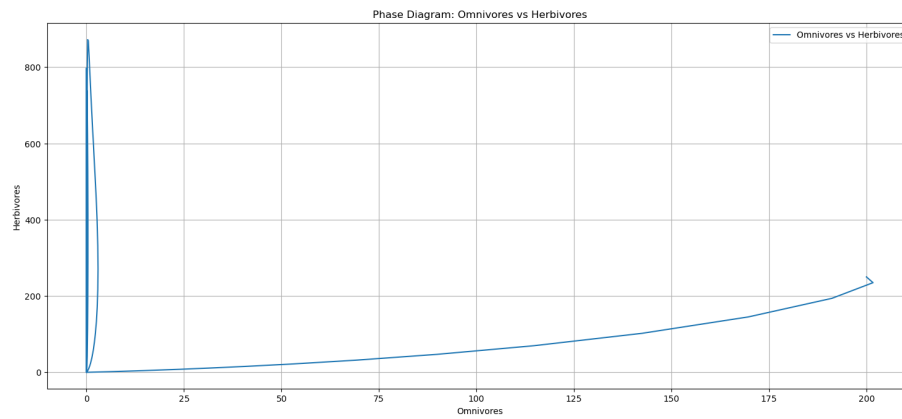
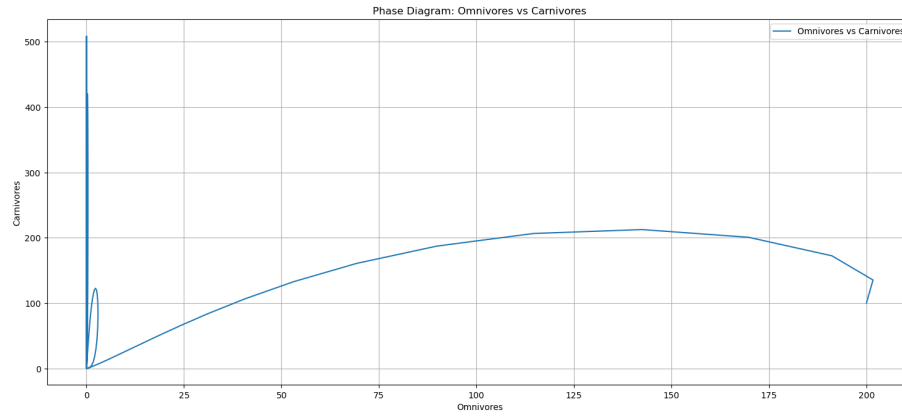


Figure 5: Phase Diagram: Herbivores vs Carnivores

The phase diagram of these two shows similarity as they are cyclical in pattern and it reflects a typical predator-prey relationship. Herbivores thrive when plants are abundant, followed by a decline as plants decrease due to consumption. Similarly, carnivores grow as herbivore population increases leading to a peak and subsequent decline in herbivores.

3.1.2 Omnivores vs Carnivores — Omnivores vs Herbivores — Decomposers vs Saprophages — Scavengers vs Saprophages — Plants vs Carnivores



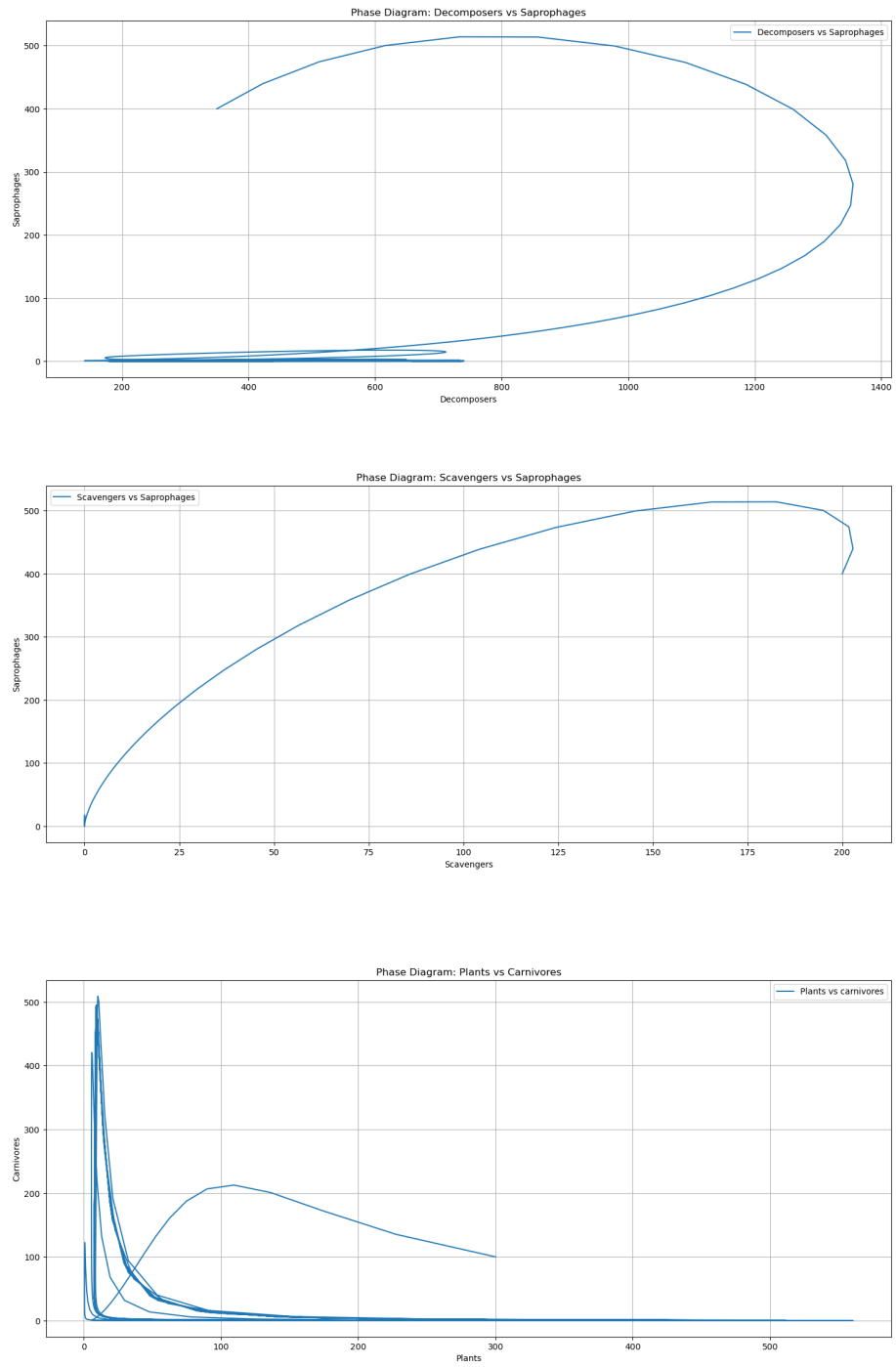


Figure 6: Five Experimental Phase Diagrams

The rest of the phase diagrams collectively highlight the interconnection and feedback mechanisms in this complex ecosystem. Each pair of species demonstrates predator-prey or competitive relationships, leading to oscillatory behavior driven by resource availability and competition.

4 Conclusion

After thoroughly analyzing this model, we shed new light on the importance of all species of ecosystem. We try to emphasize the need for biodiversity, complex relationships, and nutrient recycling. We note that these seven species are interdependent directly or indirectly. If one of them is removed, the ecosystem will be disrupted. Because we do not have real-time data, we are unable to build an accurate Phase Diagram.

5 Acknowledgements

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